

CS471 – Lesson 23 Solutions Key

Temporal Reasoning: Hidden Markov Models

Part 1: Guided In-Class Worksheet (Instructor-Led)

1. Observable vs. Hidden State Architectures

- a. **Question:** Explain the fundamental difference between a standard Markov Chain and a Hidden Markov Model (HMM) in terms of state visibility.

Solution

In a standard **Markov Chain**, the system state X_t is directly and completely observable at every time step. In a **Hidden Markov Model (HMM)**, the true state X_t cannot be directly perceived; the agent only receives indirect, noisy sensor observations (emissions) E_t generated probabilistically according to $P(E_t | X_t)$.

- b. **Question:** State the two conditional independence assumptions enforced by the graph topology of an HMM:

Solution

- **State Transition Assumption:** $P(X_t | X_{0:t-1}, E_{1:t-1}) = P(X_t | X_{t-1})$ (The current hidden state depends strictly on the previous hidden state).
- **Sensor Emission Assumption:** $P(E_t | X_{0:t}, E_{1:t-1}) = P(E_t | X_t)$ (The current evidence emission depends strictly on the current hidden state).

2. Two-Step Forward Filtering Mechanics

Scenario: $B(X_0) = \langle P(S) = 0.80, P(T) = 0.20 \rangle$, observation $E_1 = H$.

- a. **Question:** Compute the time-projected belief distribution $\bar{B}(X_1) = \langle \bar{B}(S), \bar{B}(T) \rangle$.

Solution

$$\bar{B}(S) = P(S | S)B(S_0) + P(S | T)B(T_0) = (0.70)(0.80) + (0.10)(0.20) = 0.56 + 0.02 = \mathbf{0.58}$$

$$\bar{B}(T) = P(T | S)B(S_0) + P(T | T)B(T_0) = (0.30)(0.80) + (0.90)(0.20) = 0.24 + 0.18 = \mathbf{0.42}$$

$$\bar{B}(X_1) = \langle 0.58, 0.42 \rangle$$

- b. **Question:** Compute unnormalized joint probabilities, marginal evidence, and the normalized belief state $B(X_1)$ given $E_1 = H$.

Solution

Unnormalized Probabilities:

$$P(X_1 = S, E_1 = H) = P(H | S)\bar{B}(S) = 0.20 \times 0.58 = \mathbf{0.116}$$

$$P(X_1 = T, E_1 = H) = P(H | T)\bar{B}(T) = 0.90 \times 0.42 = \mathbf{0.378}$$

Marginal Evidence Probability:

$$P(E_1 = H) = 0.116 + 0.378 = \mathbf{0.494}$$

Normalized Filtered Belief:

$$B(S_1) = \frac{0.116}{0.494} \approx \mathbf{0.2348} \quad (23.48\%)$$

$$B(T_1) = \frac{0.378}{0.494} \approx \mathbf{0.7652} \quad (76.52\%)$$

$$B(X_1) = \langle P(\text{Search}) \approx 0.2348, P(\text{Track}) \approx 0.7652 \rangle$$

Part 2: Student Independent Practice Solutions

A. Submarine Acoustic Tracking

Scenario: $B(X_0) = \langle P(B) = 0.60, P(D) = 0.40 \rangle$, observation $E_1 = L$.

1. **Question:** Compute the predicted belief state $\bar{B}(X_1)$.

Solution

$$\bar{B}(B) = P(B | B)B(B_0) + P(B | D)B(D_0) = (0.80)(0.60) + (0.30)(0.40) = 0.48 + 0.12 = \mathbf{0.60}$$

$$\bar{B}(D) = P(D | B)B(B_0) + P(D | D)B(D_0) = (0.20)(0.60) + (0.70)(0.40) = 0.12 + 0.28 = \mathbf{0.40}$$

$$\bar{B}(X_1) = \langle 0.60, 0.40 \rangle$$

2. **Question:** Compute the unnormalized likelihood-weighted probabilities given $E_1 = L$.

Solution

$$P(X_1 = B, E_1 = L) = P(L | B)\bar{B}(B) = 0.10 \times 0.60 = \mathbf{0.060}$$

$$P(X_1 = D, E_1 = L) = P(L | D)\bar{B}(D) = 0.80 \times 0.40 = \mathbf{0.320}$$

$$\text{Total Evidence: } P(E_1 = L) = 0.060 + 0.320 = \mathbf{0.380}.$$

3. **Question:** Compute the final normalized filtered belief $B(X_1)$.

Solution

$$B(B_1) = \frac{0.060}{0.380} = \frac{3}{19} \approx \mathbf{0.1579} \quad (15.79\%)$$

$$B(D_1) = \frac{0.320}{0.380} = \frac{16}{19} \approx \mathbf{0.8421} \quad (84.21\%)$$

$$B(X_1) = \langle P(\text{Battery}) \approx 0.1579, P(\text{Diesel}) \approx 0.8421 \rangle$$

B. Inferential Task Classification

Solution

- Filtering** ($P(X_t | e_{1:t})$): Estimating current state in real time).
- Decoding** ($\arg \max_{x_{1:t}} P(x_{1:t} | e_{1:t})$): Finding the most likely global sequence of states).
- Smoothing** ($P(X_k | e_{1:t})$ for $k < t$): Reconstructing past states using past and future evidence).